Baseline readiness (full-environment recovery rehearsal)

Northwind Logistics (fictional) - Scenario SCN-BASELINE - Incident 2026-10-20 09:00 UTC - Data as of 2026-10-20 09:00 UTC
Data source: Bundled synthetic baseline - Analysis 2026.10.1 - Generated 2026-10-28 07:09 UTC

1. Executive summary

- Recovery readiness is 85.7/100 (Moderate). 8 of 13 systems have evidence of a recent successful end-to-end restore test.
- The recovery plan covers 13 system(s). Estimated time to restore: At least 1 d 7 h (durations unknown for: hr-portal); 1 system(s) cannot be restored.
- Potential data-loss exposure for systems being restored: 10 h; no usable recovery point for aws-s3-exports (loss not bounded).
- 1 critical and 6 high-severity findings were raised. 3 system(s) would lose more data than their RPO allows; 3 plan step(s) miss their RTO.
- Blocked: aws-s3-exports. These cannot be restored with the current evidence and dependencies.

Metric	Value	Note
Recovery readiness	85.7 / 100	Moderate
Restore-tested systems	8 / 13	evidence-backed end-to-end tests
Stale or missing recovery points	3	RPO breached or no eligible point
Integrity failures	1	recovery points failing verification
Unmet RTO	3	1 unknown
Unmet RPO	3	1 unknown
Dependency failures	0	blocked by prerequisites, cycles or gaps
Business capability impacted	0%	criticality-weighted

2. Incident description

No incident is injected. Every protected asset is assessed as if the whole environment had to be restored from backup at the inventory's as-of time.

3. Assessment scope and asset impact

13 protected systems assessed. Plan scope: all protected systems (full-recovery rehearsal). Business capability impacted: 0% (criticality-weighted).

4. Recovery readiness, backup findings and RTO/RPO compliance

System	Tier	Evidence	Selected point	Data loss	RPO	Ready	RTO	Score
Core DNS & Network Services	tier0	Restore-tested	RP-DNS-202610200200-P	7 h	met (1 d)	T+15 min	met (30 min)	100
Identity Provider & SSO	tier0	Restore-tested	RP-IDP-202610200830-P	30 min	met (1 h)	T+1 h	met (1 h)	100
Secrets & Key Management	tier0	Restore-tested	RP-KMS-202610200800-P	1 h	met (4 h)	T+40 min	met (45 min)	100
Orders Database (PostgreSQL)	tier0	Restore-tested	RP-DBORD-202610200845-P	15 min	met (15 min)	T+1 h 55 min	met (2 h)	100
Customer Database (PostgreSQL)	tier0	Integrity only	RP-DBCUS-202610200800-P	1 h	met (1 h)	T+2 h 15 min	breached (2 h)	66
Orders API Service	tier0	Restore-tested	RP-ORDAPI-202610200200-P	7 h	met (1 d)	T+2 h 20 min	met (2 h 30 min)	100
Customer Storefront	tier0	Restore-tested	RP-WEB-202610200200-P	7 h	met (1 d)	T+2 h 50 min	met (3 h)	100
Internal ERP	tier1	Restore-tested	RP-ERP-202610200400-P	5 h	breached (4 h)	T+3 h 55 min	met (8 h)	70
Corporate File Storage	tier2	Integrity only	RP-FILES-202610200100-P	8 h	met (1 d)	T+1 d 7 h	breached (1 d)	56
Central Logging & SIEM	tier1	Restore-tested	RP-SIEM-202610200700-P	2 h	breached (1 h)	T+1 h 45 min	met (4 h)	80

System	Tier	Evidence	Selected point	Data loss	RPO	Ready	RTO	Score
Monitoring & Alerting	tier2	Integrity only	RP-MON-202610200000-P	9 h	met (1 d)	T+2 h	met (8 h)	76
HR Self-Service Portal	tier3	Integrity only	RP-HR-202610192300-P	10 h	unknown (undoc.)	n/a	unknown (undoc.)	36
Analytics Export Bucket (AWS S3)	tier3	No usable point	none eligible	not bounded	breached (1 d)	blocked	breached (2 d)	15

Score = criticality-weighted mean of per-system scores: backup confidence 35, RPO 20, RTO in a full-recovery rehearsal 20, dependencies 15, isolation 10. Unknown values earn no credit.

5. Recovery dependency order

Estimated recovery time: At least 1 d 7 h (durations unknown for: hr-portal); 1 system(s) cannot be restored. Potential data-loss exposure: 10 h; no usable recovery point for aws-s3-exports (loss not bounded).

#	Wave	System	Prerequisites	Recovery point	Start	Ready	Basis	RTO
1	1	Core DNS & Network Services	-	RP-DNS-202610200200-P	T+0 min	T+15 min	measured (EV-RST-DNS-202610031000)	met
2	2	Secrets & Key Management	dns-core	RP-KMS-202610200800-P	T+15 min	T+40 min	measured (EV-RST-KMS-202608301100)	met
3	2	Identity Provider & SSO	dns-core	RP-IDP-202610200830-P	T+15 min	T+1 h	measured (EV-RST-IDP-202609281400)	met
4	3	Customer Database (PostgreSQL)	dns-core, kms-secrets	RP-DBCUS-202610200800-P	T+40 min	T+2 h 15 min	measured (EV-RST-DBCUS-202606151300)	breached
5	3	Orders Database (PostgreSQL)	dns-core, kms-secrets	RP-DBORD-202610200845-P	T+40 min	T+1 h 55 min	measured (EV-RST-DBORD-202610120900)	met
6	4	Orders API Service	db-orders, idp-core, kms-secrets	RP-ORDAPI-202610200200-P	T+1 h 55 min	T+2 h 20 min	measured (EV-RST-ORDAPI-202610051200)	met
7	5	Customer Storefront	app-orders-api, db-customers, idp-core	RP-WEB-202610200200-P	T+2 h 20 min	T+2 h 50 min	measured (EV-RST-WEB-202610051500)	met
8	2	Central Logging & SIEM	dns-core	RP-SIEM-202610200700-P	T+15 min	T+1 h 45 min	measured (EV-RST-SIEM-202609201000)	met
9	4	Internal ERP	db-orders, idp-core	RP-ERP-202610200400-P	T+1 h 55 min	T+3 h 55 min	measured (EV-RST-ERP-202609101600)	met
10	3	Monitoring & Alerting	dns-core, idp-core	RP-MON-202610200000-P	T+1 h	T+2 h	estimated	met
11	3	Corporate File Storage	idp-core	RP-FILES-202610200100-P	T+1 h	T+1 d 7 h	estimated	breached
12	-	Analytics Export Bucket (AWS S3)	-	-	BLOCKED		no eligible recovery point	breached
13	4	HR Self-Service Portal	file-share, idp-core	RP-HR-202610192300-P	T+1 d 7 h	T+Unknown	unknown	unknown

- Recovery starts when the incident is declared (T+0). Detection, triage, forensic hold and decision time are not included in estimates.
- The parallel estimate assumes unlimited restore capacity, so independent assets restore simultaneously; the serial estimate assumes one asset at a time. Reality lies between the two.
- Restore durations come from the most recent passing restore test where one was measured; otherwise the documented estimate is used. Unknown durations are not guessed.
- An asset may start restoring only after every prerequisite in the plan has finished.
- Assets outside the plan scope are assumed to remain operational.

6. Findings and remediation recommendations

[CRITICAL] Estimated recovery exceeds RTO - Customer Database (PostgreSQL)

F-RTO-001-db-customers | rule RTO-001 | basis: derived | priority 1

Customer Database (PostgreSQL) is estimated to be ready at T+2 h 15 min, beyond its RTO of 2 h.

Calculation: wait for prerequisites 40 min + restore 1 h 35 min (measured (EV-RST-DBCUS-202606151300)) = 2 h 15 min; RTO 2 h; over by 15 min.

Business impact: Tier 0 - Mission critical. 1 dependent system(s): Customer Storefront.

Action: Shorten the recovery path: pre-stage restore infrastructure, parallelise dependency restores, or renegotiate the RTO with the business owner. Verify by: Measure the restore duration in a test and recompute the recovery plan. Expected outcome: Estimated ready time is within the RTO.

[HIGH] Unapproved roles can delete recovery points - Primary Backup Vault

F-ACC-001-vault-primary | rule ACC-001 | basis: verified | priority 2

Unapproved role(s) can delete recovery points in Primary Backup Vault: platform-admin. Approved: backup-admin.

Business impact: Tier 0 - Mission critical.

Action: Reduce backup-administrator privileges: remove delete permissions from unapproved roles and require multi-party approval for deletion. Verify by: Perform an access review and record it as evidence. Expected outcome: Only approved roles can delete recovery points.

Evidence: EV-ACC-PRIMARY

[HIGH] Contradictory evidence - Identity Provider & SSO

F-EV-002-idp-core-RP-IDP-202610200730-P | rule EV-002 | basis: verified | priority 3

RP-IDP-202610200730-P: reported integrity check status contradicts evidence; treated as 'failed'.

Business impact: Tier 0 - Mission critical. 6 dependent system(s): Internal ERP, Orders API Service, Corporate File Storage, HR Self-Service Portal, Monitoring & Alerting, Customer Storefront.

Action: Investigate the contradiction between the backup platform and the evidence store; re-run verification. Verify by: Re-verify the recovery point and confirm reported status and evidence agree. Expected outcome: Reported status and evidence agree.

Evidence: EV-INT-IDP-202610200730-P

[HIGH] No recent end-to-end restore test - Customer Database (PostgreSQL)

F-RP-005-db-customers | rule RP-005 | basis: verified | priority 4

No passing end-to-end restore test within 90 days for Customer Database (PostgreSQL) (last passing test 2026-06-15 13:00 UTC, 126 days before assessment).

Business impact: Tier 0 - Mission critical. 1 dependent system(s): Customer Storefront.

Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

Evidence: EV-RST-DBCUS-202606151300

[HIGH] No verified isolated recovery copy - Internal ERP

F-ISO-001-app-erp | rule ISO-001 | basis: verified | priority 5

Tier 1 - Business critical assets require an isolated, immutable copy; isolation status is 'none'.

Business impact: Tier 1 - Business critical.

Action: Create an isolated backup copy in a separate account/tenant with compliance-mode immutability. Verify by: Confirm a recovery point exists in the isolated repository and that its immutability configuration is evidenced. Expected outcome: Isolation status becomes 'verified'.

[HIGH] Recovery point objective breached - Internal ERP

F-RP-002-app-erp | rule RP-002 | basis: derived | priority 6

Newest eligible recovery point for Internal ERP is 5 h old at the time of the incident; RPO is 4 h.

Calculation: Incident 2026-10-20 09:00 UTC - eligible point RP-ERP-202610200400-P 2026-10-20 04:00 UTC = 5 h exposure; RPO 4 h; gap 1 h.

Business impact: Tier 1 - Business critical. Potential data loss of 5 h exceeds tolerance by 1 h.

Action: Correct the backup schedule or failing jobs so that the backup interval is shorter than the RPO, and alert on missed backups. Verify by: Review backup job logs for the last 7 days and confirm a verified recovery point exists within the RPO window. Expected outcome: Data-loss exposure is at or below the RPO.

[HIGH] Recovery point objective breached - Central Logging & SIEM

F-RP-002-logging-siem | rule RP-002 | basis: derived | priority 7

Newest eligible recovery point for Central Logging & SIEM is 2 h old at the time of the incident; RPO is 1 h.

Calculation: Incident 2026-10-20 09:00 UTC - eligible point RP-SIEM-202610200700-P 2026-10-20 07:00 UTC = 2 h exposure; RPO 1 h; gap 1 h.

Business impact: Tier 1 - Business critical. Potential data loss of 2 h exceeds tolerance by 1 h.

Action: Correct the backup schedule or failing jobs so that the backup interval is shorter than the RPO, and alert on missed backups.

Verify by: Review backup job logs for the last 7 days and confirm a verified recovery point exists within the RPO window. Expected outcome: Data-loss exposure is at or below the RPO.

[MEDIUM] Retention lock can be bypassed by privileged roles - Primary Backup Vault

F-ISO-003-vault-primary | rule ISO-003 | basis: verified | priority 8

Governance-mode retention on Primary Backup Vault can be bypassed by: backup-admin, platform-admin.

Business impact: Tier 0 - Mission critical.

Action: Move critical recovery points to compliance-mode retention or remove bypass permissions from day-to-day administrator roles.

Verify by: Review the repository policy and confirm bypass permissions are restricted to a break-glass role. Expected outcome: No standing role can bypass retention.

Evidence: EV-IMM-PRIMARY

[MEDIUM] Retention below policy minimum - Customer Database (PostgreSQL)

F-RET-001-db-customers | rule RET-001 | basis: verified | priority 9

Retention is 14 days; policy minimum for tier0 is 35 days.

Calculation: 14 d < 35 d required

Business impact: Tier 0 - Mission critical. 1 dependent system(s): Customer Storefront.

Action: Improve recovery-point retention to meet the policy minimum. Verify by: Confirm the retention setting and the oldest available recovery point. Expected outcome: Retention meets policy minimum.

[MEDIUM] Status claimed without supporting evidence - Central Logging & SIEM

F-EV-001-logging-siem-RP-SIEM-202610200800-P | rule EV-001 | basis: unknown | priority 10

RP-SIEM-202610200800-P: backup platform reports the integrity check as passed, but no evidence record supports it.

Business impact: Tier 1 - Business critical.

Action: Investigate missing backup evidence and ensure verification results are exported to the evidence store. Verify by: Locate or regenerate the evidence record and link it to the recovery point. Expected outcome: Claim is backed by an evidence record.

[MEDIUM] Newest recovery point integrity not verified - Internal ERP

F-RP-004-app-erp | rule RP-004 | basis: unknown | priority 11

Newest recovery point RP-ERP-202610200800-P (2026-10-20 08:00 UTC) has integrity status 'not_run' and is not counted as recoverable.

Business impact: Tier 1 - Business critical.

Action: Enable automated integrity verification (checksum/test mount) on every recovery point and store the results as evidence. Verify by: Confirm an integrity-check evidence record with result 'pass' exists for the newest point. Expected outcome: Newest point is integrity-verified and selected for recovery.

[MEDIUM] Newest recovery point integrity not verified - Central Logging & SIEM

F-RP-004-logging-siem | rule RP-004 | basis: unknown | priority 12

Newest recovery point RP-SIEM-202610200800-P (2026-10-20 08:00 UTC) has integrity status 'unknown' and is not counted as recoverable.

Business impact: Tier 1 - Business critical.

Action: Enable automated integrity verification (checksum/test mount) on every recovery point and store the results as evidence. Verify by: Confirm an integrity-check evidence record with result 'pass' exists for the newest point. Expected outcome: Newest point is integrity-verified and selected for recovery.

[MEDIUM] No recent end-to-end restore test - Corporate File Storage

F-RP-005-file-share | rule RP-005 | basis: unknown | priority 13

No passing end-to-end restore test within 90 days for Corporate File Storage (no passing restore test on record).
Business impact: Tier 2 - Important. 1 dependent system(s): HR Self-Service Portal.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[MEDIUM] No recent end-to-end restore test - Monitoring & Alerting

F-RP-005-monitoring | rule RP-005 | basis: unknown | priority 14

No passing end-to-end restore test within 90 days for Monitoring & Alerting (no passing restore test on record).
Business impact: Tier 2 - Important.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[MEDIUM] Estimated recovery exceeds RTO - Corporate File Storage

F-RTO-001-file-share | rule RTO-001 | basis: derived | priority 15

Corporate File Storage is estimated to be ready at T+1 d 7 h, beyond its RTO of 1 d.
Calculation: wait for prerequisites 1 h + restore 1 d 6 h (estimated) = 1 d 7 h; RTO 1 d; over by 7 h.
Business impact: Tier 2 - Important. 1 dependent system(s): HR Self-Service Portal.
Action: Shorten the recovery path: pre-stage restore infrastructure, parallelise dependency restores, or renegotiate the RTO with the business owner. Verify by: Measure the restore duration in a test and recompute the recovery plan. Expected outcome: Estimated ready time is within the RTO.

[MEDIUM] No usable recovery point - Analytics Export Bucket (AWS S3)

F-RP-001-aws-s3-exports | rule RP-001 | basis: unknown | priority 16

Analytics Export Bucket (AWS S3) has no backup policy and no recovery points.
Business impact: Tier 3 - Supporting.
Action: Establish a working backup for this asset and produce at least one integrity-verified recovery point in an isolated repository. Verify by: Run a backup job, record a passing integrity check, and confirm the point appears as eligible in RecoveryProof. Expected outcome: Asset shows an eligible recovery point and RP-001 is no longer raised.

[LOW] Backup encryption absent or undocumented - Corporate File Storage

F-ENC-001-file-share | rule ENC-001 | basis: unknown | priority 17

Backup encryption is not documented for confidential data.
Business impact: Tier 2 - Important. 1 dependent system(s): HR Self-Service Portal.
Action: Enable encryption at rest for backups using a managed key with restricted access, and document the configuration. Verify by: Record an encryption-configuration evidence record. Expected outcome: Backup encryption is enabled and evidenced.

[LOW] No recent end-to-end restore test - HR Self-Service Portal

F-RP-005-hr-portal | rule RP-005 | basis: unknown | priority 18

No passing end-to-end restore test within 90 days for HR Self-Service Portal (no passing restore test on record).
Business impact: Tier 3 - Supporting.
Action: Schedule and perform an end-to-end restore test into an isolated environment, including application-level validation. Verify by: Record the restore test result and measured duration as evidence. Expected outcome: Asset recovery evidence becomes 'Restore-tested' with a measured duration.

[LOW] RTO or RPO not documented - HR Self-Service Portal

F-RTO-002-hr-portal | rule RTO-002 | basis: unknown | priority 19

RTO and RPO not documented for HR Self-Service Portal.
Business impact: Tier 3 - Supporting.
Action: Document a missing RTO and RPO with the business owner. Verify by: Confirm RTO and RPO values are recorded in the inventory. Expected outcome: RTO/RPO compliance can be evaluated.

[LOW] Restore duration unknown - HR Self-Service Portal

F-RTO-003-hr-portal | rule RTO-003 | basis: unknown | priority 20

No measured or estimated restore duration for HR Self-Service Portal.

Business impact: Tier 3 - Supporting.

Action: Perform a restore test to measure the restore duration. Verify by: Record a restore test with a measured duration. Expected outcome: Restore duration is measured.

7. Evidence references

EV-ACC-PRIMARY, EV-IMM-PRIMARY, EV-INT-IDP-202610200730-P, EV-RST-DBCUS-202606151300

8. Unverified assumptions

- Analytics Export Bucket (AWS S3): restore duration unknown.
- Corporate File Storage: restore duration 1 d 6 h is an estimate, not a measurement.
- HR Self-Service Portal: RTO/RPO undocumented; compliance not assessed.
- HR Self-Service Portal: ready time cannot be computed (unknown duration upstream or undocumented RTO).
- HR Self-Service Portal: restore duration unknown.
- Monitoring & Alerting: restore duration 1 h is an estimate, not a measurement.
- All infrastructure, backup and evidence data in this analysis is synthetic unless the data source states otherwise.
- A recovery point counts as recoverable only when an evidence record shows a passing integrity check; reported status without evidence is treated as unknown.
- Restore-tested means a passing end-to-end restore test exists within the policy window; it validates the procedure, not every individual recovery point.
- Unknown values earn no score credit and are reported as 'Unknown' rather than assumed.
- Data-loss exposure is the time between the incident and the newest eligible recovery point; transactions in flight and replication lag are not modelled.
- All timestamps are UTC.

9. Limitations and confidence

Moderate confidence: 8/13 systems have restore-tested evidence; 1 RPO and 1 RTO values could not be evaluated. Report generated 2026-10-28 07:09 UTC from analysis version 2026.10.1.

- RecoveryProof is a decision-support tool. It does not perform, guarantee or replace tested enterprise backup and recovery.
- Conclusions are only as good as the evidence supplied; missing evidence is reported as unknown, never as success.
- Time estimates exclude detection, triage, forensic preservation and business validation, and assume restore capacity as stated in the plan assumptions.
- The local audit trail is tamper-evident (hash-chained), not tamper-proof.